1.1 THE FUNDAMENTAL THEOREM OF ALGEBRA

In this book, a polynomial

$$f(z) = a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + a_n z^n \tag{1.1.1}$$

is always supposed to have complex coefficients $a_0, \ldots, a_n$ unless otherwise indicated. If $a_n \neq 0$, then the polynomial f is of *degree n*. This is written as $\deg f = n$, and a_n is called the *leading coefficient.* A polynomial with leading coefficient 1 is said to be *monic.*

An important relation between two polynomials f and g, with $\deg f = n$ and $\deg g = m \leq n$, is the *division transformation.* It guarantees a representation

$$f(z) = q(z)g(z) - r(z) \tag{1.1.2}$$

where q is a polynomial of degree $n-m$ and r is one of degree less than m. A constructive verification of (1.1.2) may be carried out by equating coefficients or

by a division scheme. Repeating this process with the pair f, g replaced by g, r, etc., we arrive at the *Euclidean algorithm.* It may formally be described by

$$\begin{aligned} f_0 &:= f(z), \qquad f_1(z) := g(z), \\ f_j(z) &=: q_j(z) f_{j+1}(z) - f_{j+2}(z) \qquad (j = 0, 1, \ldots). \end{aligned} \tag{1.1.3}$$

Since the degrees of the polynomials f_j are strictly decreasing, we finish with an equation $f_{k-1}(z) = q_{k-1}(z) f_k(z)$, where $f_k(z) \not\equiv 0$. We may also allow the possibility that $\deg g > \deg f$. Then (1.1.2) holds with $q(z)$ being identically zero and $r(z) = -f(z)$, that is, $f_2 = -f$. Hence, except for a sign, the first step of the Euclidean algorithm interchanges f and g. The relations (1.1.3) show that any common divisor of f_0 and f_1 is a divisor of all the successors $f_2, \ldots, f_k$. Conversely, f_k is a divisor of all its predecessors $f_{k-1}, \ldots, f_0$. Hence f_k is a *greatest common divisor* of f and g. Moreover, we have factorizations

$$f(z) = f_k(z)\varphi(z) \quad \text{and} \quad g(z) = f_k(z)\psi(z),$$

where φ and ψ are appropriate polynomials.

A point $\zeta \in \mathbb{C}$ is a *zero* of *multiplicity* m, or of *order* m, or an *m-fold* zero of f if

$$f(\zeta) = f'(\zeta) = \cdots = f^{(m-1)}(\zeta) = 0 \quad \text{and} \quad f^{(m)}(\zeta) \neq 0.$$

A zero of multiplicity 1 is said to be *simple.* If $f(\zeta) \neq 0$, then we may consider ζ as a zero of multiplicity 0. In the case where $m \geq 1$, the division transformation with $g(z) := (z-\zeta)^m$ yields that $f(z) = q(z)(z-\zeta)^m$. In fact, if the polynomial r in (1.1.2) were not identically zero, then either r itself or one of its derivatives of order less than m would be a non-vanishing constant. However, $r^{(\mu)}(\zeta) = 0$ for $\mu = 0, \ldots, m-1$; this is a contradiction.

In this way we also conclude that a polynomial of degree $n \geq 1$ cannot have more than n zeros, counted according to their multiplicities. However, it is a non-trivial question whether an arbitrary polynomial of positive degree has a zero at all. Its answer is crucial for the whole book, and is the main result of this section.

Theorem 1.1.1 (Fundamental theorem of algebra) *Every polynomial of positive degree has a complex zero.*

Proof Let f be a polynomial of positive degree. Then $|f(z)|$ tends to infinity as $|z| \to \infty$. Hence there exists an $R > 0$ such that $|f(z)| > |f(0)|$ for $|z| > R$. As a continuous function, $|f(\cdot)|$ attains a minimum on the compact disc $\overline{\mathcal{D}}(0; R)$. If this occurs at ζ, then $|f(\zeta)| \leq |f(0)|$, and so $|f(\zeta)| = \min_{z \in \mathbb{C}} |f(z)|$.

Now, assuming that f has no zero in $\mathbb{C}$, we may write

$$f(\zeta + z) = a + bz^k + z^{k+1} g(z)$$

with some integer $k \geq 1$, coefficients $a, b \in \mathbb{C} \setminus \{0\}$, and a polynomial $g(z)$. Corresponding to ω, chosen as a k-th root of $-a/b$, there exists a $t \in (0,1)$ so that $t \left|\omega^{k+1} g(t\omega)\right| < |a|$. For this pair t, ω, we obtain

$$\begin{aligned} |f(\zeta + t\omega)| = \left|a + b(t\omega)^k + (t\omega)^{k+1} g(t\omega)\right| &= \left|a(1-t^k) + (t\omega^{k+1} g(t\omega))t^k\right| \\ &< |a|(1-t^k) + |a|\, t^k = |a| = |f(\zeta)|. \end{aligned}$$

This is a contradiction. □

We already know from the division transformation that, if ζ is a zero of f, then $f(z) = (z - \zeta)f_1(z)$, where $\deg f_1 = \deg f - 1$. Repeating this reasoning, we may restate Theorem 1.1.1 in the following refined form.

Theorem 1.1.2 *Every polynomial f of positive degree n has a total of n complex zeros, and may be represented as*

$$f(z) = a_n \prod_{j=1}^{k} (z - z_j)^{m_j},$$

with $a_n \neq 0$, distinct zeros $z_1, \ldots, z_k$, and positive integers $m_1, \ldots, m_k$ satisfying $m_1 + \cdots + m_k = n$. □

1.8 NOTES

Section 1.1

The Euclidean algorithm is named in appreciation of an analogous division scheme used by Euclid (about 300 BC) for the determination of the greatest common divisor of two integers. Descartes observed in 1637 that, if ζ is a zero of $f(z)$, then $z - \zeta$ factors out; probably, this was already known to Thomas Harriot (about 1560–1621).

Possibly the first approach to the fundamental theorem of algebra was made by Peter Roth when he mentioned in his *Arithmetica philosophica* (Nuremberg, 1608) that an algebraic equation can have as many solutions as its degree, but not more. He added (freely translated from German) that this does not mean that every cubic, biquadratic, quintic, etc. equation must have three, four, five, etc. solutions, but that the maximum number of solutions is equal to the degree. We do not know if he had thought of unsolvable equations, or only on a reduction of the number of *distinct* solutions due to multiple roots.

The next step was made by A. Girard in his *L'invention nouvelle en l'algebre* of 1629 where he stated without a proof: 'Toutes les equations d'algebre recoivent autant de solutions, que la denomination de la plus haute quantité le demonstre, excepté les incomplettes.' Later, he did not mention exceptions any more, but gave explanations which amount to the observation that a solution can be real, complex, or 'something similar'.

First attempts for a proof of the fundamental theorem of algebra were made by d'Alembert in 1746, Euler in 1749, Lagrange in 1772, and Laplace in 1795. Gauss criticized all of them. D'Alembert came to a situation where he had to assume the existence of a minimum; it would now follow by a compactness argument, which was not available in his time. The major objection against the three other proofs was that their authors already assumed the existence of some obscure solution (as Girard had postulated) and proved that it must be a complex number. Again, these days, this cannot be seen as a severe problem since now it is not difficult to show by purely algebraic means that for every polynomial with coefficients in a field there exists a splitting field. Thus algebra guarantees that a polynomial with complex coefficients has certain 'generalized' zeros, while the fundamental theorem of algebra states that they are again complex numbers, or equivalently, that *the field of complex numbers is algebraically closed.*

It should be emphasized that, despite its name, the fundamental theorem of algebra is not a purely algebraic theorem. Since the complex numbers form a topological field, proofs of that theorem cannot avoid analytical, topological, or geometrical tools completely, although they may sometimes be quite hidden.

Gauss himself gave four proofs, one in 1799, two in 1816, and one in 1849; see (Gauss 1876, pp. 1–30, 31–56, 57–64, 71–102). The first proof is contained in his doctoral dissertation, written at the age of 22. He considered polynomials $f(z)$ with real coefficients and justified, by delicate geometric arguments, that the curves $\Re f(z) = 0$ and $\Im f(z) = 0$ intersect. His fourth proof is of a similar nature, but he allowed complex coefficients. Analyzing these two proofs, Ostrowski in 1920 pointed out some possible gaps and presented the desirable supplements in a rigorous form, see (Gauss 1981, Part 2, article 3, pp. 1–18; Ostrowski 1983, pp. 538–53). An incontestable variant of Gauss's first proof was also given by Motzkin and Ostrowski (1933).

Gauss's second proof is strongly algebraic. He adopted ideas of Euler, but did calculations in terms of an unknown instead of assuming the existence of a 'generalized' zero. It is in a similar way that the existence of a splitting field is proved these days. His third proof is purely analytic, and is related to a familiar proof in complex analysis using a contour integral of f'/f.

For more details and references on the history of the fundamental theorem of algebra, see (Ebbinghaus *et al.* 1983, Chapter 4; Fine and Rosenberger 1997). Up to now, an overwhelming number of proofs have appeared. We mention only a few, classifying them according to their authors' major intention.

Elementary proofs are those which do not need more than standard calculus. The above-mentioned proof of d'Alembert is of that type. It was simplified and improved by Argand (1814) and Cauchy (1820). More modern versions are due to Fefferman (1967) and Terkelsen (1976), whose proof is the one presented in this book. Other elementary proofs were obtained by Redheffer (1957, 1964) and Wolfenstein (1967).

Complex analysis proofs are amongst the shortest and most familiar ones. Gauss's third proof belongs to this class. Bieberbach (1952) presents five different proofs: one by Ankeny (1947), who used Cauchy's integral theorem directly; a proof by van der Waerden based on the maximum modulus principle; the familiar proofs using the theorems of Liouville and Rouché; and Cauchy's proof, which Bieberbach based on the open mapping theorem. Ankeny's proof was modified by Boas (1964). The principle of the argument and a related formula for the contour integral of a logarithmic derivative lead to further proofs.

Predominantly algebraic proofs. The most prominent representatives are the proof of Laplace in 1795 and the second proof of Gauss. The ingenuity of Laplace's proof in a completed form is exhibited in (Ebbinghaus *et al.* 1983, pp. 95–7). It needs four tools, of which three are purely algebraic, while the only analytic tool is the intermediate-value theorem, used to guarantee that a polynomial of odd degree with real coefficients has a real zero. For references to further strongly algebraic proofs, see (Ebbinghaus *et al.* 1983).

Predominantly real proofs aim at avoiding complex numbers. Gauss's first proof is of that type. For other such contributions, see (Perron 1949), whose proof is also *elementary*, and the above-mentioned papers of Redheffer. By considering $f(z)\overline{f(\overline{z})}$ instead of $f(z)$, it is enough (though not easier) to give a proof for polynomials with real coefficients.

Topological proofs. Arnold (1949) uses Brouwer's fixed point theorem, while Stein (1954) has a short proof based on elementary homology theory.

Constructive proofs were already proposed by Weierstrass in 1859 and 1891 (Weierstrass 1894, pp. 247–56; 1903, pp. 251–69). They have to satisfy further criteria for being acceptable as 'intuitionistic' (Brouwer and de Loor 1924; Weyl 1924, pp. 142–6; H. Kneser 1940; Rosenbloom 1945; M. Kneser 1981). H. W. Kuhn (1974) gave an unusual constructive proof based on a labelling procedure and a combinatorial lemma. In recent times, never-failing root-finding algorithms have gained a lot of attention, e.g., (Dejon and Nickel 1969; Hirsch and Smale 1979; Smale 1981; Ruscheweyh 1984). They can serve as constructive proofs provided that, after a careful study of their foundations, possible non-constructive arguments or assumptions concerning the existence of zeros have been eliminated.

Theorem 1.1.3, in its full generality, is proved in any book on complex analysis, e.g., (Ahlfors 1966, p. 151). Theorem 1.1.4 is due to Montel (1934/35).